MILITARY SPECIFICATION

RECEIVER, RADIO R-1041B/ARN

1. SCOPE

1.1 This document covers one type of 75 Megacycle (MC) Aircraft marker Beacon Receiver for one light operation, designated Receiver, Radio R-1041B/ARN. Mounting MT-2292/AR is used with the receiver.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 — Box, Fiberboard.
PPP-B-640 — Boxes Corrugated, Triple Wall, 350 Pounds Maximum Weight.
PPP-F-320 — Fiberboard, Corrugated and Solid, Sheet Stock Container Grade and Cut Shapes.
PPP-S-760 — Strapping, Non Metallic, and Connectors.
PPP-T-76 — Tape Pressure-Sensitive Adhesive Paper, Water Resistant (Carton Sealing).
PPP-T-97 — Tape: Pressure-Sensitive Adhesive Packaging and Packing of FSC 5825.

MILITARY

MIL-P-116 — Preservation, Methods of.
MIL-T-5422 — Testing Environmental Aircraft Electronic Equipment.
MIL-I-6181 — Interference Control Requirements Aircraft.
MIL-M-13231 — Marking of Electronic Items.
MIL-F-14072 — Finishes for Ground Signal Equipment.
MIL-B-43291 — Box, Fiberboard, Corrugated, Double-Wall Weather Resistant.
MIL-P-55110 — Printed Wiring Boards.

STANDARDS

FEDERAL

FED STD-595 — Colors

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 — Marking for Shipment and Storage.
MIL-STD-147 — Palletized Unit Loads (40 x 48 in. 4-Way Partial and 4 Way Pallets).

FSC 5826

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MIL-STD-169 — Extreme Temperature Cycle.

MIL-STD-170 — Moisture Resistance Test Cycle for Ground Signal Equipment.

MIL-STD-252 — Wired Equipment, Classification of Visual and Mechanical Defects.

MIL-STD-704 — Electric Power, Aircraft Characteristics and Utilization of.

DRAWINGS

ELECTRONIC COMMAND

SC-DL-425200-E — Receiver, Radio R-1041B/ARN.

SC-DL-425260-E — Bracket, Mounting MT-2292/AR.

(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, 1424 16th Street, N.W., Washington, D.C.)

ASSOCIATION OF AMERICAN RAILROADS

Uniform Freight Classification

(Application for copies should be addressed to the Association of American Railroads, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 **Description.** Receiver, Radio R-1041-B/ARN is a completely transistorized 75 megacycle fixed frequency aircraft marker beacon set.

3.2 **Contruction.** The equipment shall be constructed in accordance with the following drawing and data lists:

SC-DL-425200-E — Receiver, Radio R-1041B/ARN.

SC-DL-425260-E — Mounting MT-2-292/AR.

3.3 **Preproduction samples.** The contractor shall furnish two preproduction samples for approval, if required by the invitation for bids and contract (see 4.3 and 6.2d).

3.4 **Finish, protective.** The equipment shall be given a protective finish in accordance with MIL-F-14072 and the equipment drawings. The final paint film shall be color No. 37038 (lusterless aviation black) according to Federal Standard No. 595 (see 4.4).

3.5 **Marking.**

3.5.1 *General.* Marking shall conform to MIL-M-13231 and the equipment drawings (see 4.4). Front panel markings shall be Group I as described in that specification.

3.5.2 *Visibility.* Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with a minimum disassembly of the equipment.

3.5.3 *Serial numbers.* Radio Receiver R-1041B/ARN shall be serial numbered.

3.6 **Printed wiring assemblies.** Printed wiring assemblies shall be in accordance with MIL-P-55110 and the equipment drawings (see 4.4).

3.7 **Riveting.** After riveting, the joints shall be tight, the joined parts shall be properly seated and tight against their bearing surfaces.

3.8 **Service conditions.** The equipment shall meet the following service conditions:

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3.8.1 *Temperature.* The equipment shall be subjected to the temperature cycle of MIL-STD-169 modified as indicated in 3.8.1.1 (see 4.7.1).

3.8.1.1 *Operating.* Ambient temperature in the range of +125° to —40°F. (The +125°F temperature shall include effect of sun-load.) Exposure at the high temperature not to exceed 4 hours, and at the low temperature not to exceed 72 hours at any one time.

3.8.1.2 *Nonoperating.* Exposure in the range of -160°F to —80°F. Exposure at high temperature not to exceed 4 hours, and at low temperature not to exceed 24 hours at one time.

3.8.2 *Relative humidity.* Exposure to 97 percent relative humidity for 20 hours; and exposure at 100 percent humidity, with condensation, for 4 hours for each 24 hours over a 10 day cycle (see 4.7.2).

3.8.3 *Altitude* (see 4.7.3).

3.8.3.1 *Operating.* Altitude conditions up to 10,000 feet above sea level.

3.8.3.2 *Nonoperating.* Up to 50,000 feet above sea level.

3.8.4 *Vibration and shock.* The equipment shall suffer no physical damage and shall meet full specification requirements when subjected to the vibration and shock tests of 4.7.4, 4.7.5 and 4.7.5.1.

3.9 Electrical requirements.

3.9.1 *Time delay characteristics.* With an RF input from 1000 microvolts + 1, —2 db to 350,000 microvolts, the indicator lamp shall instantly follow the modulation when keyed at the rate of 6 pulses per second, each pulse lasting 1/15 second (see 4.6.3).

3.9.2 *Audio output.* Minimum audio output into a 150 ohm load impedance shall be at least 20 milliwatts when a 5000 microvolt standard test signal of 75 megacycle frequency, amplitude modulated 90 percent by 1300 cps, produced by a signal generator having an internal impedance of 52 ohms is fed to the input of an antenna simulator (see 4.6.4).

3.9.3 *Audio response.* The audio response shall not vary more than ± 5db at the high and low temperature extremes; and not vary more than ± 3 db at ambient room temperature, from reference obtained at 1300 cps when audio frequency is varied from 380 to 3150 cps (see 4.6.5).

3.9.4 *Automatic gain control.* The automatic gain control circuit shall prevent the Aural Power output from rising more than 6 db above the aural light-on-point when the radio frequency input is increased 100,000 microvolts. The aural light-on-point is defined as the amount of unattenuated audio produced across an 150 ohm impedance at the instant the indicator control circuit triggers the lamp indicator to the ON position (see 4.6.6).

3.9.5 *Indicator Control circuit.* The indicator control circuitry, using the audio, shall trigger the indicator lamp, rated at 28 volts dc, 0.1 ampere. The light-on-point (in microvolts) for any frequency between 380 and 3150 cps shall be within ± 2db from a reference obtained by applying a standard test signal (see 4.6.7).

3.9.6 *Differential.* A radio-frequency voltage reduction of not more than 15 percent from the voltage required to turn on the indicator light shall turn off the lamp. Differential is defined as the difference between the RF voltages at the light-on-point and at the light-off point expressed as percentages of the light-on-point RF voltage. In the event the lamp flickers at light-on-point, as RF increase of 3 db or more shall cause the lamp to remain on (4.6.8)

3.9.7 *Power source and power consumption.* The receiver shall operate from any nominal 28 volt dc power source. Power consumption of the R-1041B ARN shall not exceed 55 milliamperes (exclusive of the indi-

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cator lamp current) at 28 volts dc when measured under any natural combination of service conditions specified herein. The received shall not be damaged due to inadvertent reversing of the power source polarity (see 4.6.9).

3.9.8 *Power source voltage variation.* The receiver sensitivity (voltage at the input to the antenna simulator required to trigger the lamp indicator and at the same time produce 5 to 20 milliwatts of unattenuated audio across an 150 ohm load when a test signal, modulated by the desired frequency, is applied) shall not differ by more than 6 db from a reference taken at 28 volts when the power source input voltage is varied from 42 to 32 volts (see 4.6.10).

3.9.9 *Power source transients.* Operation of the receiver shall not be affected when subjected to transient conditions of MIL-STD-704 (see 4.6.11).

3.9.10 *Supply voltage ripple.* With no signal input, a voltage of 1.5 RMS at any frequency between 200 and 5000 cps present on the dc supply in put line, shall not cause the audio output to increase by more than 0.6 milliwatt or activate the indicator light (see 4.6.12).

3.9.11 *Selectivity.* The overall selectivity of the receiver shall be not less than ± 40 kilocycles from the center frequency at 6 db down points (total bandwidth of at least 80 kilocycles). The 60 db down points shall be no greater than ± 150 kilocycles from center frequency. (Center frequency is defined as 75.000 megacycles; see 4.6.13.)

3.9.12 *Sensitivity.* The sensitivity of the receiver shall be variable between 500 and 5000 microvolts in both the HI and LOW sensitivity modes of operation with a standard test signal applied to the antenna simulator. Except during test of above sensitivity range requirements, the LOW sensitivity potentiometer R21 shall be set for standard sensitivity of 300 microvolts ± 1 db, and the HI sensitivity potentiometer R59 shall be set for standard sensitivity of 1000 microvolts ± 1 db, and the potentiometer adjustments shall not be changed except as required by the test procedure (see 4.6.14).

3.9.13 *Stability.* The light-on-point shall not differ by more than 4.5 db when the radio frequency is varied ± 30 kc from a reference light-on-point taken at 75 megacycles (see 4.6.15).

3.9.14 *Signal-plus-noise-to-noise.* At light-on-point the receiver signal-plus-noise-to-noise ratio shall be 20 db or greater (see 4.6.16).

3.9.15 *Spurious response.* The response to any frequency from 0.2 to 1000 megacycles shall be down 60 db, or greater, relative to the response at 75 megacycles, except the response at 66.6 megacycles and 76.0 megacycles shall be attenuated 50 db minimum relative to the response at 75 megacycles (see 4.6.17).

3.9.16 *Volume attenuation.* The volume attenuation switch shall attenuate the audio output from 0 db to 21 ± 30 db in four steps of 5 ± 1.5 db (see 4.6.18).

3.9.17 *Remote variable control.* A remote variable control potentiometer with a resistance of 150 ohms, and having an "ON"-"OFF" switch shall control the audio output. The HI or LOW sensitivity shall be remotely selected by means of the single pole single throw switch (see 4.6.19).

3.9.18 *Electro-magnetic interference.* The equipment shall meet the requirements of MIL-I-6181 (see 4.6.20).

3.9.19 *Operation.* Each receiver will be subjected to the operational test of 4.6.21.

3.10 **Technical literature, tools, and running spare parts.** Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in

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the equipment furnished on the order (see 6.2(c) and 6.2(f)).

3.11 **Workmanship.** The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs (see 4.8):

In this Specification —
3.2 Construction
3.7 Riveting

4. QUALITY ASSURANCE PROVISIONS.

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Classification of inspection.** Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery) (see 4.3).
(b) Quality conformance inspection.
(1) Acceptance inspection of equipment before preparation for delivery (see 4.4 and 4.5).
(2) Acceptance inspection of preparation for delivery (see 4.9).

4.3 **Preproduction inspection.** This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I, the inspection specified in the subsidiary documents covering items listed in 4.4, and the inspection specified for Group A, Group B, and Group C (see tables II, III, and IV respectively). The preproduction inspection will normally be performed in this order: (1) Vibration, (2) Moisture resistance, (3) Shock, bench handling, or other preproduction inspection may precede, follow, or be interspersed between the foregoing.

TABLE I. *Preproduction Inspection*

Inspection	*Rqt. para*	*Test para*
Vibration	3.8.4	4.7.4
Humidity	3.8.2	4.7.2
Shock	3.8.4	4.7.5
Electro-magnetic Interference	3.9.18	4.6.20

4.4 **Acceptance inspection covered by subsidiary documents.** The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery.

Item	*Where Required*
Finish	3.4
Marking	3.5
Printed Wiring Assemblies	3.6

4.5 **Acceptance inspection of equipment before preparation for delivery.** The contractor to demonstrate compliance with specified requirements shall perform the inspections specified in 4.4 through 4.5.3.

This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractors inspection procedures and examine the contractor's inspection records. In addition the Government at its descretion, may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 *Group A inspection.* This inspection, including sampling, shall conform to Table II and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational test shall be last.

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TABLE II. *Group A inspection*

Inspection	Rqr. Para	Test Para	Major	Minor
VISUAL AND MECHANICAL	3.11	4.8		
Receiver R-1041B/ARN prior to application of conformal coating after application of conformal coating			2.5% 1.0%	6.5% 4.0%
ELECTRICAL				
Time delay characteristic	3.9.1	4.6.3		
Audio output	3.9.2	4.6.4		
Audio response	3.9.3	4.6.5		
AGC	3.9.4	4.6.6		
Indicator control circuit	3.9.5	4.6.7		
Differential	3.9.6	4.6.8		
Power source and power consumption	3.9.7	4.6.9		
Power sourve voltage variation	3.9.8	4.6.10		
Selectivity	3.9.11	4.6.13	2.5 DPHU*	
Stability	3.9.13	4.6.15		
Signal and noise to noise	3.9.14	4.6.16		
Volume attenuation	3.9.16	4.6.18		
Remote control	3.9.17	4.6.19		
Overall operation	3.9.19	4.6.21		

DPHU = Defects per hundred units
* All electrical defects are considered major.

4.5.2 *Group B inspection.* This inspection including sampling, shall conform to table III and to special procedures for small-sample inspection of MIL-STD-105. The reduced inspection procedures shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected on units that have been subjected to and met the Group A inspection.

4.5.2.1 *Group B sampling plans.* The Group B sampling plans, for the AQL's listed in table III, shall be as follows:

AQL	*Inspection level for normal and tightened inspection*	*Inspection level for reduced inspection*
6.5%	S-7	S-3

4.5.2.2 *Order of inspection within Group B.* Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE III. *Group B inspection*

Inspection	Rqr Para	Insp Para
Power source transient	3.9.9	4.6.11
Supply voltage ripple	3.9.10	4.6.12
Spurious response	3.9.15	4.6.17

4.5.3 *Group C inspection.* This inspection shall be as listed in table IV, and shall normally be performed on sample units which have been subject to and met Group A and B inspection. Two units, for each Group C inspection, unless otherwise noted in table IV, shall be selected each month without regard to their quality except that the units inspected at the start of the contract shall be selected from the first units produced. No defects shall be allowed.

TABLE IV. *Group C inspection*

Inspection	Rqr para	Test para
Temperature	3.8.1	4.7.1
Altitude	3.8.3	4.7.3
Shock, Bench handling	3.8.4	4.7.5.1
Vibration	3.8.4	4.7.4
Electro-Magnetic Interference	3.9.18	4.6.20

* To be performed on one sample each third month (90 days).

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4.5.3.1 *Noncompliance.* If a sample unit fails Group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same condition, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.5.4 *Reinspection of conforming Group B and Group C sample units.* Unless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection or both, may be accepted on contract, provided they are resubjected to and pass Group A inspection after repair of all visible damage.

4.6 Electrical tests.

4.6.1 *Test Standards.*

4.6.1.1 *Standard test signal.* The standard test signal is a 75 megacycle radio frequency signal, amplitude modulated 90 percent by 1300 cps, produced by a signal generator having an internal impedance of 52 ohms. This signal is fed to the input of an antenna simulator.

4.6.1.2 *Aural light-on-point.* The aural light-on-point is the amount of unattenuated audio produced across a 150 ohm impedance at the instant the indicator control circuit triggers the lamp indicator to the ON position.

4.6.1.3 *Differential.* Differential is the difference between the RF voltages at the light-on-point and the light-off-point expressed as a percentage of the light-on-point RF voltage. The indicator lamp is on, even if the light flickers.

4.6.1.4 *Light-on-point.* The light-on-point is defined as the amount of voltage present at the input to the antenna simulator the instant the indicator lamp goes on. The indicator lamp is on even if the light flickers.

4.6.1.5 *Light-off-point.* The light-off-point is defined as the amount of voltage present at the input to the antenna simulator the instant the indicator lamp goes off and there is no indication of lamp flicker.

4.6.1.6 *Sensitivity.* Sensitivity is measured at the antenna simulator input, and is defined as the voltage required to trigger the lamp indicator and at the same time produce 5 to 20 milliwatts of unattenuated audio across a 150 ohm load when a test signal, modulated by the desired audio frequency, is applied.

4.6.1.7 *Antenna simulator.* The antenna simulator is a 6 db, 50 ohm attenuator (Stoddart Aircraft Radio Company Inc. or equal) inserted between the signal generator output and the antenna input of the receiver.

4.6.2 *Initial test set-up.* The receiver shall be tested using a test set-up similar to that shown in figure 1. Interconnect the R-1041B/ARN and the associated test equipment as shown in figure 2. The signal generator shall have been in operation at least 15 minutes before any tests are made on the equipment. Adjust the direct current voltage supply to 28 volts at the input of the bench test harness and proceed in accordance with the following:

(1) Calibrate the Hewlett-Packard 608D signal generator, or equal, to a frequency of 75 megacycles.
(2) Adjust the Hewlett-Packard 200 AB oscillator, or equal, to 1300 cycles per second and indicate 90 percent modulation on the 608 D signal generator.
(3) Set General Radio output meter 583A, or equal, for a 150 ohm output load.
(4) Set Switch SW-1 (meter-headset) on test set-up in the "meter" position.

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(5) Set volume control on the R-1041B/ARN to maximum.
(6) Switch test set-up to "ON" position.

This test set-up shall be used in all receiver performance tests specified herein unless modified in the individual test paragraph.

4.6.3 *Time delay characteristics test.* Insert a relay micro-positioner between the audio oscillator and the signal generator and utilize function generator Hewlett Packard 202B, or equal, to operate the relay. Set the receiver sensitivity to the high position and feed a standard test signal with keyed modulation to the receiver. Control the function generator to operate the relay to deliver 3 pulses and pause 6 pulses at a rate of 6 pulses per second, each pulse lasting 1/15 second. Observe and record indicator lamp characteristics while increasing RF input from 0 to 350,000 microvolts. Change modulation frequency to 400 cps and 3000 cps and repeat the above procedures, recording results. The requirements of 3.9.1 shall be met.

4.6.4 *Audio output test.* Increase RF output of the signal generator and record voltage present when 20 milliwatts is indicated on the output meter. The requirements of 3.9.2 shall be met.

4.6.5 *Audio response test.* Apply the standard test signal and increase RF output until a reference of 5 milliwatts is indicated on the output meter. Change the audio frequency, keeping the modulation at 90 percent, to 385 cps, 400 cps, 500 cps, 800 cps, 1300 cps, 2090 cps, 2500 cps, 3000 cps, and 3150 cps and record audio output. The requirements of 3.9.3 shall be met.

4.6.6 *Automatic gain control test.* Set the sensitivity control to the high position. Record audio output for values of RF input from 1000 microvolts through 100,000 microvolts at 6 db intervals of RF input. The requirement of 3.9.4. shall be met.

4.6.7 *Indicator control circuit test.* Apply the standard test signal and set the receiver sensitivity switch to the high position. Record RF-light-on-point and audio output and R-F light-off-point and audio output. Change modulation frequency to 380 cps and record RF light-on-point and audio output and RF light-off-point and audio output. Change modulation frequency to 3150 cps and record light-on-point and audio output, and RF light-off-point, and audio output. Compute in db the difference in RF light-on-point for each modulation frequency using the standard test signal as referenced. Compute differential for each light-on-point versus light-off-point. In the event lamp flickers at light-on-point, note condition and increase RF input and record, in db, the voltage increase required to cause the lamp to remain on. The requirements of 3.9.5 shall be met.

4.6.8 *Differential test.* The difference between the RF voltages at the light-on-point and at the light-off-point obtained in the Indicator Control Circuit Test (4.6.7), shall meet the requirement of 3.9.6.

4.6.9 *Power source and power consumption test.* The receiver shall be checked for compliance with the requirements of 3.9.7.

4.6.10 *Power source voltage variation test.* Set supply voltage to equipment for 28 volts dc and sensitivity switch to the high position. Apply a standard test signal and record light-on-point and audio output. Repeat this procedure for 24, 29, and 32 volts. Compute the difference, in db, in the light-on-points due to variation of the supply voltage. The requirements of 3.9.5 shall be met.

4.6.11 *Power source transient test.* Connect the R-1041B/ARN to the dc power source through a one henry choke. Charge a 30 microfarad capacitor with a 80 volt power supply. Parallel a 10,000 ohm resistor across the dc supply line. Discharge the 30 mfd capacitor across the resistor and observe the transient voltage on a Tectronix ± 545, Oscilloscope, or equal. Record sensitivity measurement before and after this test. The requirements of 3.9.9 shall be met.

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4.6.12 *Supply voltage ripple test.* Use the test instrumentation as described in 4.6.2. with the exception that a modulation transformer and an additional audio oscillator, Hewlett Packard, ±200 CD, or equal, shall be utilized to introduce a 1.5 volt ac ripple into the dc supply. Set the receiver at high sensitivity and record light-on-point and audio output, with and without the ripple on the dc supply line. Set the receiver at light-on-point and vary the ripple frequency from 200 to 5000 cps. Note any change in light-on-point or audio output in reference to frequency of ripple voltage. Remove the standard test signal (RF) while introducing 1.5 volts ac on the dc supply line and record, with reference to frequency of ripple, audio output and note any indication of the indicator lamp. The requirements of 3.9.10. shall be met.

4.6.13 *Selectivity test.* Set the receiver controls for standard HI sensitivity (1000 microvolts - 1 db) operation and the audio attenuation switch for minimum output. Using a standard test signal, increase the RF input signal until the receiver audio output meter indicates a reference output of one milliwatt. Record this reference RF input and then double the RF input for the 6 db points. Increase the frequency of the signal generator until the audio output decreases to the one milliwatt reference output level and record the frequency. Decrease the frequency of the signal generator through the resonant point, until the audio output is again one milliwatt, and record the frequency. Compute and record the bandwidth for the 6 db down points and total bandwidth. Repeat the above for the 60 db down points (1000 times the reference RF input) (see 3.9.11).

4.6.14 *Sensitivity range tests.* Adjust the HI-LOW sensitivity control for LOW sensitivity operation. Apply the standard test signal. Then measure and record the receiver sensitivity with LOW sensitivity potentiometer R 21 adjusted to jack of the test set-up (see figure 2) and switch SW-1 to headset position. Increase RF output of signal generator to maximum audio signal to headset. Rotate volume control on test set-up until audio signal is completely attenuated. The requirements of 3.9.12 shall be met.

4.6.14.1 *Standard sensitivity test.* Adjust the HI-LOW sensitivity control for HI sensitivity operation, then for LOW sensitivity operation. Apply a standard test signal and measure and record the receiver standard sensitivity requirement of 3.9.12.

4.6.15 *Stability test.* Using the standard test signal (75 MC) as a reference, record light-on-point and audio output. Reduce frequency of the signal generator to 74.970 megacycles and record light-on-point and audio output. Increase frequency of the signal generator to 75.030 megacycles and record light-on-point and audio output. Compute in db plus and minus 30 kilocycles light-on-point from reference 75 megacycles signal. The requirements of 3.9.13 shall be met.

4.6.16 *Signal-plus-noise-to-noise test.* Set receiver sensitivity control to HI position. Apply the standard test signal and note audio output at light-on-point. Remove modulation from standard test signal and record audio output (noise). Repeat procedure with the sensitivity control as the LOW position. The requirements of 3.9.14 shall be met.

4.6.17 *Spurious response.* Using the standard test signal as a reference, increase RF output until one milliwatt is indicated on the output meter. Change signal generator to 200 kilocycles and increase RF output by 60 db. Sweep signal generator from 200 kilocycles to 1000 megacycles and record any response less than 60 db down relative to the response at 75 megacycles. Requirements of 3.9.15. shall be met.

4.6.18 *Volume attenuation test.* Increase signal generator RF voltage to obtain maximum audio output and record milliwatt output and reading of db scale on the output meter. Decrease step attenuator to minimum audio output and record readings for each step (4 steps). Requirements of 3.9.16 shall be met.

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4.6.19 *Remote control test.* Insert high impedance headset into audio jack of the test set-up (see figure 2) and switch SW-1 to headset position. Increase RF output of signal generator to maximum audio signal to headset. Rotate volume control on test set-up until audio signal is completely attenuated. The requirements of 3.9.17 shall be met.

4.6.20 *Electro-magnetic interference.* The receiver shall be tested in accordance with MIL-I-6181.

4.6.21 *Operation.* Operate the equipment continuously for a period of 20 hours with the sensitivity switch in the HI position and with a standard test signal of 2000 microvolts applied. During each hour the receiver shall be subjected to ten minutes of keyed modulation. This keyed modulation shall be performed using the same test set up and pulse durations as for the time delay characteristic test (see 4.6.3).

4.7 Service tests.

4.7.1 *Temperature.* The equipment shall be subjected to the temperature cycle shown on MIL-STD-169 modified as indicated in 3.8.1.1. The equipment temperature shall be allowed to stabilize before testing. Sensitivity controls (R59 and R21) shall not be readjusted during the test. Equipment shall be tested at steps 4 and 8 of the temperature cycle and shall meet the requirements of 3.9.1, 3.9.12, and 3.9.13, except that the requirements of 3.9.12 are for standard HI and LOW sensitivity requirements only, which are degraded to 1000 microvolts ± 6 db and 3000 microvolts ± 6 db and the requirements of 3.9.1 are degraded to 1000 microvolts ± 6 db. The equipment shall be tested at step 10 of the temperature cycle and shall meet the electrical tests of Group A inspection (see 4.5.1, table II).

4.7.2 *Relative humidity.*

4.7.2.1 *Test method.* The equipment shall be tested as follows:

(a) Precondition equipment dry at 135° ± 5°F for 24 hours.

(b) Condition at 77° ± 5F at 40 to 50 percent relative humidity for 24 hours.

(c) Subject the equipment to continuous moisture resistance cycling for five 48 hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL-STD-170. During the test, the equipment shall be placed in the test chamber in a manner similar to that in which it will be used in service. Provisions shall be made to prevent the dripping of water on to the equipment from above. Moisture shall be provided by steam or by evaporation of tapwater having a pH value between 6.5 and 7.5 measured at 25°C. The velocity of air throughout the exposure area shall not exceed 150 feet per minute. The test chamber shall be vented to the atmosphere to prevent the buildup of pressure.

(d) After the cycling has been completed, the final condition period shall begin. This shall consist of conditioning the equipment for 24 hours at 77° ± 5°F and 40 to 60 percent relative humidity.

4.7.2.2 *Performance.*

(a) During the test measurement periods specified on the standard, the equipment shall be tested, and meet the requirements specified in 3.9.6 and 3.9.12 with exception that the reference light-on-point limit is increased 6 db and the requirements in 3.9.6 are increased 10 percent to a total of 25 percent.

(b) After the completion of the final conditioning period, the equipment shall be tested and meet the performance requirements of the

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electrical tests of Group A inspection see 4.5.1, table II).

4.7.3 *Altitude.* The equipment placed in its normal operating position in an altitude chamber, shall be tested as follows:

4.7.3.1 With the equipment in the chamber, operating stabilized at room temperature, the receiver shall be tested and meet the following requirements:

3.9.1 Time delay
3.9.12 Sensitivity (Standard only)
3.9.13 Stability

4.7.3.2 Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulating 10,000 feet above sea level) and after temperature and pressure stabilization, repeat the measurements specified in 4.7.3.1. The equipment shall meet specified performance for those measurements.

4.7.3.3 Remove the power from the equipment and further reduce the barometric pressure to 3.4 ± 0.1 inches of mercury (simulating 50,000 feet above sea level). Maintain this condition for one hour.

4.7.3.4 Return the chamber to ambient pressure. Apply power to the equipment and after stabilization, repeat the measurements of 4.7.3.1. The equipment shall meet specified performance for these measurements.

4.7.4 *Vibration.* The equipment shall be tested to Curve IV of figure 2 in MIL-T-5422. During the test the equipment shall be tested and meet the performance requirements listed in 4.7.3.1. The equipment, after vibration, shall meet the performance requirements of the electrical tests of the Group A inspection (see 4.5.1, table II).

4.7.5 *Shock.* The equipment shall be subjected to the shock tests of MIL-T-5422. Upon completion of the tests, the equipment shall be tested and meet performance requirements of the electrical tests of the Group A inspection (see 4.5.1, table II).

4.7.5.1 *Shock, bench handling.* The chassis and front panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid 2 inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axis for a total of 4 drops.

(b) Repeat (a) with the assembly resting on other faces, until it has been dropped a total of 4 times on each face on which the assembly could be practically placed for servicing. Tests shall be performed before, and after shock to insure compliance with the performance requirements listed in 4.7.3.

4.8 **Visual and mechanical inspection.** The equipment shall be examined for the defects listed in MIL-STD-252 and table V.

TABLE V. *Classification of defect*

Class	Type of defect	Classification of defect
Conformal coating	Major	Type or thickness of coating not as specified.
Conformal coating	Major	Coating does not completely cover all printed wiring.
Conformal coating	Major	Holes, cracks, chips, or crazing in coating with exposure of printed wiring.
Conformal coating	Major	Visible contamination metallic matter trapped in coating.
Conformal coating	Major	Ground contacts covered with coating, preventing good electrical ground.
Conformal coating	Minor	Visible contamination (nonmetallic foreign residue or deposits in coating).

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TABLE V. *Classification of defect* (continued)

Class	Type of defect	Classification of defect
Conformal coating	Minor	Bubbles, blisters or wrinkles.
Conformal coating	Major	Coating "tacky" (improper processing).
Conformal coating	Minor	Excessive building up of coating on, or between parts.

4.9 Acceptance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine compliance with requirement of section 5.

5. PREPARATION FOR DELIVERY.

5.1 **Preservation and Packaging.** Preservation and packaging shall be Level A or C, as specified.

5.1.1 *Level A.*

5.1.1.1 *Cleaning.* Each Receiver Radio R-1041B/ARN, shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 *Drying.* Each Receiver, Radio R-1041B/ARN, shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 *Unit packaging.* Each Receiver, Radio R-1041B/ARN, shall be packaged Method III, as follows:

Cushion each receiver on all surfaces with cells or pads, or both, fabricated of corrugated fiberboard, type CF, Class domestic, variety SW, conforming to PPP-F-320, designed to absorb the shock impact normally encountered in handling and transit. Place the cushioned receiver within a regular slotted style W6c fiberboard box, conforming to PPP-B-636. Closure shall be made in accordance with the appendix of the box specification. In addition, all seams and joints shall be sealed with pressure sensitive adhesive tape conforming to PPP-T-76.

5.1.2 *Level C.* Each Receiver, Radio R-1041B/ARN, shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity.

5.2 **Packing.** Packing shall be level A, B or C as specified (see 6.2b).

5.2.1 *Level A.*

5.2.1.1 A quantity of packaged receivers as specified in 5.1, shall be packed within a close fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant, special requirements use. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 *Palletized load.* Receivers, Radio, R-1041B/ARN, packed as specified in 5.2.1.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, Class weather resistant W5s. The load shall be bonded to the pallet by strapping.

5.2.1.3 *Less than palletized load.* When quantities per destination are less than a pallet load, a quantity of items packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box, conforming to PPP-B-640, Class 2, Grade A Style E, or MIL-B-43291. Closure shall be made in accordance with the appendix of the applicable box specification. Reinforcing shall be by non-metallic strapping, or pressure-sensitive filament tape banding conforming to PPP-S-760, type II, or PPP-T-97, type IV respectively. Selection of the material and application shall

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be in accordance with the appendix of the applicable box specification.

5.2.2 *Level B.* Receivers, Radio R-1041B/ARN, packaged as specified in 5.1, shall be packaged as specified in 5.2.1 except that the fiberboard caps shall be domestic and fiberboard boxes limited to PPP-B-636 and PPP-B-640, class domestic and class I, respectively.

5.2.3 *Level C.* Receiver, Radio R-1041B/ARN, packaged as specified in 5.1, shall be packed as specified in 5.2.1 except that pallets, fiberboard boxes and caps shall be of the type, size, and kind commonly used for the purpose and shall comply with the rules and regulations of the common carriers as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 **Marking.** In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** Receiver, Radio R-1041B/ARN mounted in an aircraft is intended for use as a single-light marker beacon receiver of ground aviation beacon signals.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this document.
(b) Level of packaging and level of packing required for shipment. (Level A, Level B, or Level C).
(c) The specific paragraphs of section 5 which are applicable to the particular procurement.
(d) Preproduction Inspection.
(e) Technical literature required.
(f) Quantity of tools and running spare parts (see 3.10).

Two samples units are generally required so that lengthy environmental tests can be completed on one sample unit while complete performance measurements can be made on the second sample unit.

6.3 **Nomenclature.** The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 **Group C Inspection** Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see table IV).

6.5 **Verification inspection.** Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality system and the quality history of the product, and will normally be identified by the category listed below:

(a) *Type A.* That inspection set forth in the Quality Assurance Provisions of this specification and the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by MIL-STD-105.
(b) *Type B.* That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of MK-STD-105.
(c) *Type C.* A reduced inspection procedure resulting in a material reduction in the amount of inspec-

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tion set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided in Type B and is based upon a consistently acceptable product resulting from a planned Quality Control System voluntarily employed by the contractor in production of the product.

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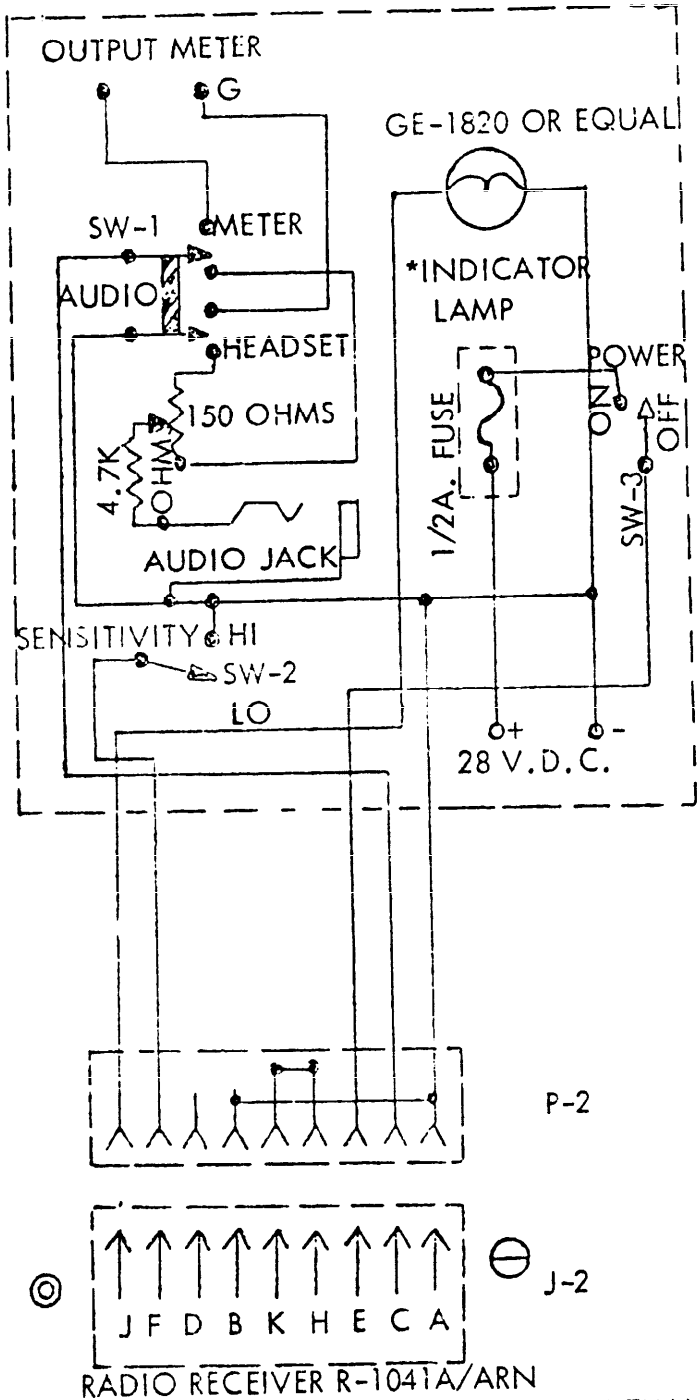
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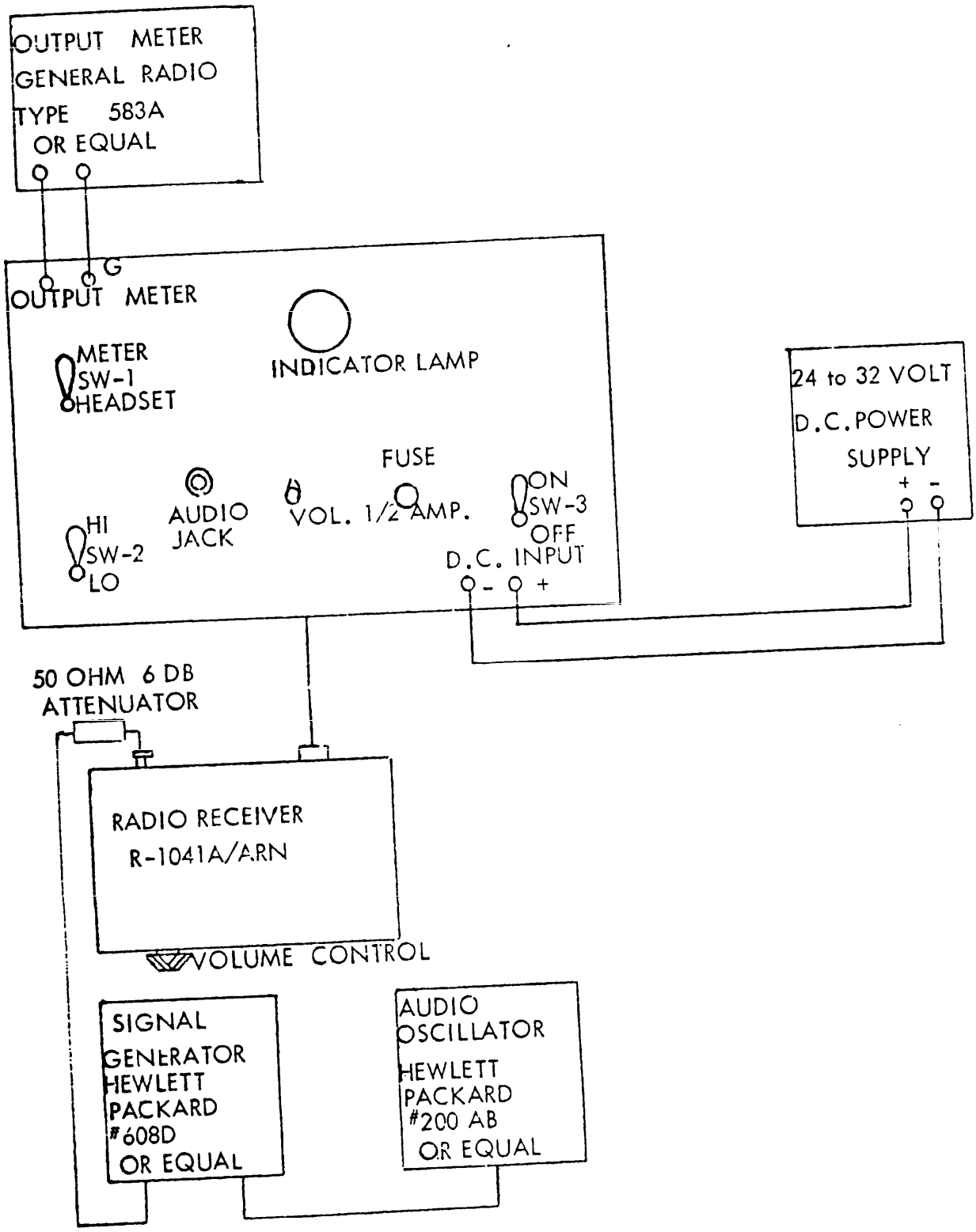


*CAUTION: USE OF A LAMP WITH GREATER CURRENT DRAIN THAN 0.1 AMPERE AT 24 VOLTS WILL RESULT IN DAMAGE TO THE EQUIPMENT.

WIRING DIAGRAM
TEST SET-UP FOR R-1041B/ARN

FIGURE 1

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BLOCK DIAGRAM

TEST SET-UP FOR R-1041B /ARN

FIGURE 2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corners, and send to preparing activity.

SPECIFICATION NO.

ORGANIZATION — CITY AND STATE

CONTRACT NO. — QUANTITY OF ITEMS PROCURED — DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

A. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

C. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

D. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF YES, IN WHAT WAY?

E. REMARKS (Any other pertinent data which may be of use in improving this specification. If there are additional papers, attach and submit with this form.)

NAME AND ACTIVITY (Printed or typed name and activity)

DATE

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